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Cartesian closedness of lax comma categories

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Abstract

We analyse cartesian closedness of the lax comma categories $Ord//X$, $Top//X$ and $Cat//X$.

This talk is based on joint work with Fernando Lucatelli Nunes, Rui Prezado e Dirk Hofmann, published at [2, 4, 1, 3].

Keywords: Lax comma categories; cartesian closed categories

Subject area: Category Theory

MSC2020: 18D15, 18A40, 18A25

References

[1] M.M. Clementino, D. Hofmann, R. Prezado, Topological lax comma categories. Order 43:3 (2026).

[2] M.M. Clementino, F. Lucatelli Nunes, Lax comma categories of ordered sets. Quaestiones Mathematicae 46 (S1) (2023) 145-159.

[3] M.M. Clementino, F. Lucatelli Nunes, Cartesian closedness of lax comma categories. DMUC Preprint 26-59. arXiv 2611.01184

[4] M.M. Clementino, F. Lucatelli Nunes, R. Prezado, Lax comma categories: cartesian closedness, extensivity, topologicity, and descent. Theory Appl. Categ. 41 (16) (2024), 516-530.